Notes on Computational Fluid Dynamics: General Principles

5.9 Systems of equations

Most CFD calculations involve solving a system of equations that represent the physics of the problem. For example, laminar flow by natural convection can be represented by the equations introduced in Sec. 2.20, reproduced below.

momentum: solve for $\mathbf{u}$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \bullet (\mathbf{u}\mathbf{u}) - \nabla \bullet (\nu \nabla \mathbf{u}) = -\nabla p + \frac{\rho(T,p)}{\rho_0}\mathbf{g}$$

mass: solve for p

$$\nabla \bullet \mathbf{u} = 0 \longrightarrow \nabla^2 p + \nabla \bullet [\nabla \bullet (\mathbf{u}\mathbf{u})] = 0$$

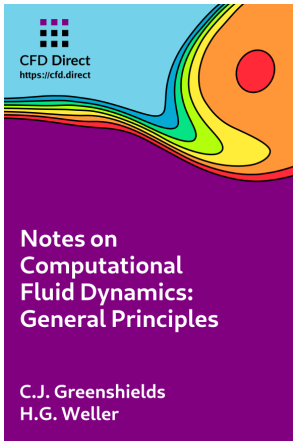
energy: solve for T

$$\frac{\partial T}{\partial t} + \nabla \bullet (\mathbf{u}T) - \nabla \bullet (\alpha \nabla T) = r$$

The system provides 3 equations (1 vector, 2 scalar) which can be solved for 3 unknowns, $\mathbf{u}$, p and T.

As discussed in Sec. 3.4, the finite volume method solves an individual matrix equation for each variable, *e.g.* $\mathbf{A} \bullet T = \mathbf{b}$ for T. The vector equation for $\mathbf{u}$ is decoupled into 3 matrix equations for individual components, *i.e.* $\mathbf{A} \bullet u_x = \mathbf{b}$, $\mathbf{A} \bullet u_y = \mathbf{b}$ and $\mathbf{A} \bullet u_z = \mathbf{b}$.

Each individual matrix equation for one solution variable, *e.g.* u_x, incorporates terms from other variables, *e.g.* ∇p, into the source vector $\mathbf{b}$. The contribution to $\mathbf{b}$ is calculated using current values of the respective variables. Systems of equations are thereby solved by *successive substitution* of solved variables into the source vectors of subsequent equations.



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An iterative solution for a single equation, like the one in Sec. 5.7, can be extended to a system of equations. Time t is incremented by Δt and equations are solved in sequence, before returning to start the next time step with the increment of t.

The substitutions in the momentum and pressure equations are particularly important, culminating in *corrections* to $\mathbf{u}$ and the advective flux ϕ_f, discussed in Sec. 5.10.

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p + \frac{\rho(T,p)}{\rho_0}\mathbf{g}$$

($\mathbf{u}\mathbf{u}$: uses ϕ_f)

$$\mathbf{A} \cdot \mathbf{u} = \mathbf{b} : \text{solve for } \mathbf{u}$$

(substitute $\mathbf{u}$)

$$\nabla^2 p + \nabla \cdot [\nabla \cdot (\mathbf{u}\mathbf{u})] = 0$$

($\mathbf{u}\mathbf{u}$: uses ϕ_f)

$$\mathbf{A} \cdot p = \mathbf{b} : \text{solve for } p$$

(substitute p to *correct* ϕ_f and $\mathbf{u}$)

At the start of any time step the current $\mathbf{u}$ becomes $\mathbf{u}^o$ for the discretisation of the $\partial \mathbf{u}/\partial t$ term in the momentum equation. The advection term $\nabla \cdot (\mathbf{u}\mathbf{u})$ is discretised by Eq. (3.8), treating one $\mathbf{u}$ as flux ϕ_f and the other as the advected quantity.

The equation is solved for $\mathbf{u}$. The new solution for $\mathbf{u}$ is substituted into the p equation which is solved for p. The new solution for p is then used to *correct* ϕ_f in order to help enforce the mass conservation constraint $\sum_f \phi_f = 0$ ($\nabla \cdot \mathbf{u} = 0$).

Before the current solution step is completed, $\mathbf{u}$ is also corrected to reduce the error in the discretisation of $\partial \mathbf{u}/\partial t$ when it then becomes $\mathbf{u}^o$ in the following solution step. The correction also provides a better "initial guess" $\mathbf{u}$ for the matrix solution of the next momentum equation, which reduces the solution time.

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